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Research Projects
I have been engaged in the following research projects:

Context Specific Search
While Web search is an essential tool for independent research, users often have to filter through results that are irrelevant, targeted at the wrong audience or from non-authoritative sources. We propose the use of search engines that are customized for each of these contexts. We automatically build such custom search engines for two very different research contexts - high school students in an AP US History class and doctors looking for information on medical conditions. We show through side by side evaluations that results from these custom search engines are preferred over results from Google by history teachers and doctors for 90% of the queries.

Publications
1. Guha, Neel, and Matt Wytock. "Course-specific search engines: semi-automated methods for identifying high quality topic-specific corpora". Proceedings of the 22nd International Conference on World Wide Web, 2013. PDF
2. Guha, Neel. "Course specific search engines: A study in incorporating context into search". Proceedings of the 22nd ACM International Conference on Information and Knowledge Management, 2013. PDF

Privacy
Third party tracking on the web, through cookies and other means, is one of the most widespread forms of user data collection. Since much can be inferred about a user by mining the pages he/she has visited (i.e. interests, gender, purchases, trips, location, etc), third parties can construct profiles of users and gather sensitive personal information. The surreptitious nature of this tracking, often occurring without the user's consent or knowledge, is alarming.

We built a browser extension, SpyWatch, (for Chrome and Firefox) that lets users 'track the trackers'. Using this we collected statistics on the extent of tracking on popular websites. Astonishingly, third party tracking occurs on the most well known and respected sites. The top ten (by traffic) online newspapers in the US have an average of 13 third party tracking sites on their homepages! We also conducted a survey of users (using this extension) to better understand attitudes towards this phenomenon, and obtained rather surprising results.

Publication
1. Guha, Neel. "SpyWatch: A Case for Transparency". Proceedings of the 12th International Semantic Web Conference, 2013. PDF

Education
We ran a pilot program during summer 2013 to explore the use of technologies such as Google Hangouts to connect students in developing countries with potential tutors across the world. Thrice a week, students from Helpers of the Handicapped, a school for handicapped orphans in Kolhapur (India), were tutored in math over Google Hangouts by high school students from the Bay Area. Using various online math games, we tried to increase students interesting in and understanding of math. In addition, the tutoring sessions attempted to increase fluency in English.

1. Guha, Neel. "Tutoring From the Desktop: Facilitating Learning Through Google+ Hangouts". Proceedings of the companion publication of the 23rd international conference on World Wide Web companion, 2014.

Applications
I have been building software tools for my community for the last three years. Some of the systems I have built include:

1. AP US History search: A search engine for students of AP US History, that avoids some of the common problems faced by students when they use Google for doing APUSH related project research. (guha.com/apushcse.html)
2. Medtree: A search engine that provides results that are targeted at medical professionals (in contrast to Google, most of whose results are targeted at patients). This was built in collaboration with doctors at Stanford Hospital. My contribution was the system that takes labeled results from doctors and generates (a much larger) list of sites to include in the search engine. In collaboration with Dr. Errol Ozdalga. (medtree.stanford.edu)
3. SpyWatch, an extension for Google Chrome that allows users to view the third party websites tracking them when they are browsing online. (install)